

Provable Quantization with Randomized Hadamard Transform

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Abstract

Vector quantization via random projection followed by scalar quantization is a fundamental primitive in machine learning, with applications ranging from similarity search to federated learning and KV cache compression. While dense random rotations yield clean theoretical guarantees, they require $\Theta(d^2)$ time. The randomized Hadamard transform HD reduces this cost to $O(d \log d)$, but its discrete structure complicates analysis and leads to weaker or purely empirical compression guarantees.

In this work, we study a variant of this approach: *dithered quantization* with a single randomized Hadamard transform. Specifically, the quantizer applies HD to the input vector and subtracts a random scalar offset before quantizing, injecting additional randomness at negligible cost. We prove that this approach is unbiased and provides mean squared error bounds that asymptotically match those achievable with truly random rotation matrices. In particular, we prove that a dithered version of TurboQuant achieves mean squared error $(\pi\sqrt{3}/2 + o(1)) \cdot 4^{-b}$ at b bits per coordinate, where the $o(1)$ term vanishes uniformly over all unit vectors and all dimensions as the number of quantization levels grows.

1 Introduction

Vector quantization is a fundamental technique for compressing real-valued vectors in $\mathbb{R}^d$ into compact binary representations. Given a vector $x \in \mathbb{R}^d$, a quantizer VECTORQUANT maps x to a sequence of $b \cdot d$ bits $\text{VECTORQUANT}(x)$. The bits can then be input to another procedure VECTORDEQUANT , obtaining the decoded vector $\tilde{x} = \text{VECTORDEQUANT}(\text{VECTORQUANT}(x))$. The size $b \cdot d$ of the compressed representation controls the distortion $\|x - \tilde{x}\|^2$, or the expected distortion $E[\|x - \tilde{x}\|^2]$ if the procedures are randomized. Alternatively, instead of decoding $\tilde{x}$, some function of it (e.g., the distance to another vector y) is estimated. This primitive plays an important role in many areas of machine learning, including similarity search [Cha02, JDS10, DIIM04, GL24, GGX⁺25], federated learning [VBP⁺22, VBBP⁺21, BBPE⁺24], efficient transformers [ZDHM25], and model compression [GKD⁺22, AMC⁺24, TCS⁺24].

A popular and well-studied approach to vector quantization is *random projection followed by scalar quantization* [Cha02, DIIM04, DCL08, AIL⁺15, VBP⁺22, VBBP⁺21, GL24, AMC⁺24, TCS⁺24, GGX⁺25, ZDHM25]. Here, one computes $\text{VECTORQUANT}(x) = \text{quant}(Rx)$, where R is a random rotation matrix and quant is a scalar quantizer (e.g., the sign function) applied coordinate-wise to the vector Rx . A key advantage of this approach is that it is *data-oblivious*: it requires no complex preprocessing tailored to the input distribution. Moreover, this method is amenable to theoretical analysis, since each coordinate of Rx is approximately normally distributed when R is a uniformly random rotation.

However, applying a dense random rotation matrix R requires $\Theta(d^2)$ time, which can be prohibitively expensive for high-dimensional vectors. To overcome this bottleneck, [AC09, Sar06, WLRT08, KW11, HMT11] and others proposed to replace random projections with *randomized Hadamard transforms*. Specifically, one replaces R with the structured matrix HD , where $H \in \mathbb{R}^{d \times d}$ is the normalized Hadamard matrix and D is a diagonal matrix with independent Rademacher (± 1) entries on the diagonal. The resulting transform can be applied in $O(d \log d)$ time using the fast Walsh-Hadamard algorithm, yielding a substantial speedup over dense rotations. Thanks to its low complexity and efficient parallel implementation, randomized Hadamard transform is often the method of choice in industrial deployments, see, e.g., [Chr25, AAH⁺24].

Unfortunately, HD is no longer a uniformly random rotation; in fact, it is not even a continuous random variable. As a consequence, the coordinates of HDx have a more complex distribution than in the dense case, and theoretical guarantees for quantization become harder to establish. Existing results are typically either empirical [AIL⁺15, SFKM17, OP25] or provable but with higher dimension or number of bits compared to those achievable with truly random rotations [AL13, KW11, Tro11, LSS14, CN22, BBPE⁺24]. A natural remedy is to use more general *structured matrices* obtained by composing multiple randomized Hadamard transforms, possibly interleaved with other (e.g., sparse diagonal) matrices [DKS11, LSS14, ZS21, KW16, ZM26]. In particular, the last paper independently showed that the composition of just *two* randomized Hadamard transforms suffices to approximate¹

uniform random rotation in high dimensions. However, those matrices are still more complex than a single randomized transform. Furthermore, the approximation error translates into an increased bound for the expected distortion after quantization.

In this paper, we study a simple alternative to composing multiple randomized Hadamard transforms, namely *dithering*. Specifically, we consider quantizers where each coordinate of HDx is shifted by a random amount (controlled by a parameter U) before applying the scalar quantizer quant.

Such quantizers are quite simple, as they require only one application of the randomized Hadamard transform, followed by a subtraction of a random vector. At the same time, the random offset injects additional randomness into the quantization process, which facilitates the analysis.

1.1 Our results

We consider the following data-oblivious vector quantization problem. For an input vector $x \in \mathbb{R}^d$, a quantizer $\text{VECTORQUANT} : \mathbb{R}^d \rightarrow \{0, 1\}^{b \cdot d}$ outputs a binary string of bd bits, where b is the bit-width per coordinate. The dequantizer $\text{VECTORDEQUANT} : \{0, 1\}^{b \cdot d} \rightarrow \mathbb{R}^d$ maps this string back to a vector $\tilde{x} \in \mathbb{R}^d$. We aim to design a pair of quantizer and dequantizer such that for any desired bit-width b minimize the following expected distortion measure for any (worst-case) vectors:

$$\sup_{x \in S^{d-1}} \mathbb{E} \|x - \tilde{x}\|_2^2,$$

where the expectation is over the internal randomness of the quantizer. The restriction to unit vectors is only for normalization: for nonzero inputs one can store the norm separately, quantize $x/\|x\|_2$, and rescale the decoded vector. In the following, we assume that d is padded to a power of 2, thus there exists a $d \times d$ Hadamard matrix.

Our main result shows that randomized Hadamard mappings HD , after dithering, provides compression guarantees that asymptotically match those achievable with truly random rotation matrices R . Moreover, our quantizer is unbiased, which is a desirable property for numerous applications. In particular we show that our proposed algorithm has the following guarantees:

Theorem 1.1. *For every dimension d and every unit vector $x \in S^{d-1}$, Algorithm 5 satisfies*

$$\mathbb{E}_{D,U}[\tilde{x}] = x$$

and

$$\mathbb{E}_{D,U} \|x - \tilde{x}\|_2^2 \leq \frac{\pi\sqrt{3}/2 + o(1)}{4^b}, \quad b \rightarrow \infty, \quad (1)$$

where the $o(1)$ term vanishes as $b \rightarrow \infty$, uniformly over d and x .

¹They show that, for each vector x and coordinate i , $(HD_1 HD_2 x)_i$ approximates $(Rx)_i$ in Kolmogorov distance up to an error of $O(d^{-1/5})$. See Related Work for further discussion.

The scaling constant in this theorem matches the bound obtained for TurboQuant in [ZDHM25] that assumed *fully random* rotation matrices, up to the $o(1)$ term.

Furthermore, as noted in [BBBIM⁺26], this also provides guarantees for a special case of the EDEN quantization method [VBP⁺22], also using fully random rotation matrices. Thus, Theorem 1.1 can be viewed as justifying the use of a single randomized Hadamard matrix in those methods.

We also provide a quantizer for inner products. Similarly to TurboQuant, we achieve this result by additionally quantizing the residual and showing that the quantization error is uncorrelated with the direction of the vector we dot product x with. Somewhat surprisingly, a classical mixed moment property of Rademacher random variables suffices to establish this fact with random rotations replaced by only a single HD matrix:

Theorem 1.2. *Let $x \in \mathcal{S}^{d-1}$ be a unit vector and let $y \in \mathbb{R}^d$. Let $\hat{x}$ be the reconstructed vector produced by Algorithm 4 with a base bit-width b . Then*

$$\mathbb{E} |\langle y, \hat{x} - x \rangle|^2 \leq 13 \left(\frac{\pi\sqrt{3}}{2} + 1 + o(1) \right) \frac{\|y\|^2}{d \cdot 4^b},$$

where the $o(1)$ term vanishes as $b \rightarrow \infty$, uniformly over d and x .

Moreover, the output of $\text{QUANTIZE}(x)$ from Algorithm 4 can be stored using $db + (3 + \frac{1}{2\ln 2})d + O(\log(b + \log d))$ total bits.

1.2 Related work

In addition to the discussion in the introduction, in this section we elaborate on connections with some prior/concurrent works in more detail.

Dithering, i.e., the use of a random offset in quantization, has a long history in quantization theory [Sch64, ZF92]. In the context of random projections combined with quantization, dithering has been employed in [DIIM04, DCL08] and more recently in [GL24]. However, their analysis assumed a truly random matrix R .

Another approach to analyzing quantization after a single randomized Hadamard transform appears in [BBPE⁺24], where the authors use the Bentkus–Dzindzalieta tail bound to compare the tails of normalized Hadamard coordinates with Gaussian tails. They then partition the range of coordinates into intervals and bound the quantization error on each interval separately, leveraging the tail bound to control contributions from the outer intervals. However, as the authors themselves note, this approach yields only a loose bound on the expected distortion.

An alternative approach to bridging the gap between a single randomized Hadamard transform and a fully random rotation is taken by a very recent independent work [ZM26], which shows that composing *two* randomized Hadamard transforms HD_1HD_2 suffices to approximate a uniform random rotation in Kolmogorov distance, with an error of $O(d^{-1/5})$ per coordinate. While this result provides a general-purpose surrogate for random rotations, the procedure is more complex, and applying it to quantization incurs an additive distortion overhead that does not vanish for fixed d .

2 Construction of the quantizer

For the ease of exposition, we separate the main mean squared error argument from the unbiasedness property. We first present a simpler baseline algorithm, Algorithm 1, which achieves the claimed mean-squared error bound (Eq. (1)) but is slightly biased. In the later Section A.3, we present Algorithm 5, which is an unbiased variant of Algorithm 1.

Our quantizer independently and randomly flips the signs of the input vector coordinates, mixes the coordinates using a normalized Hadamard transform, and then applies the same scalar quantizer to every transformed coordinate. The scalar quantizer is a dithered version of Lloyd quantization, which was also used in Turboquant [ZDHM25] and EDEN [VBP⁺22]. We define the scalar quantizer now, start by introducing some basic notations. Let

$$\varphi(t) = (2\pi)^{-1/2} e^{-t^2/2}$$

be the standard Gaussian density, and define

$$F(t) := \frac{1}{A} \int_{-\infty}^t \varphi(s)^{1/3} ds, \quad \text{where } A := \int_{\mathbb{R}} \varphi(t)^{1/3} dt.$$

Such F is the CDF of a Gaussian random variable with mean 0 and variance 3 (see Claim A.4 for the proof). In Algorithm 1, the scalar quantizer applies F to each coordinate, imposes a shifted uniform grid on the resulting quantile levels, and reconstructs each grid bucket at the inverse image of its midpoint, i.e. reconstructs each bucket at the corresponding middle quantile.

Specifically, we choose a uniformly random shift parameter $U \in [0, 1]$ and denote the parameter $B = 2^b$ (the target codebook size, i.e. number of possible quantized scalar values). Throughout this paper, we use $[B]$ to denote $\{0, 1, \dots, B-1\}$. For $j \in [B]$, we define grid points

$$h_j := (j + U)/B.$$

Define $h_0 := 0$ and $h_B := 1$. Now for $j \in [B]$ define buckets

$$\mathcal{B}_j \leftarrow \{t \in \mathbb{R} : h_j \leq F(t) < h_{j+1}\}.$$

Finally, for $j \in [B]$ define reconstruction values

$$q_j \leftarrow F^{-1}((h_j + h_{j+1})/2). \quad (2)$$

See Fig. 1 below for an illustration.

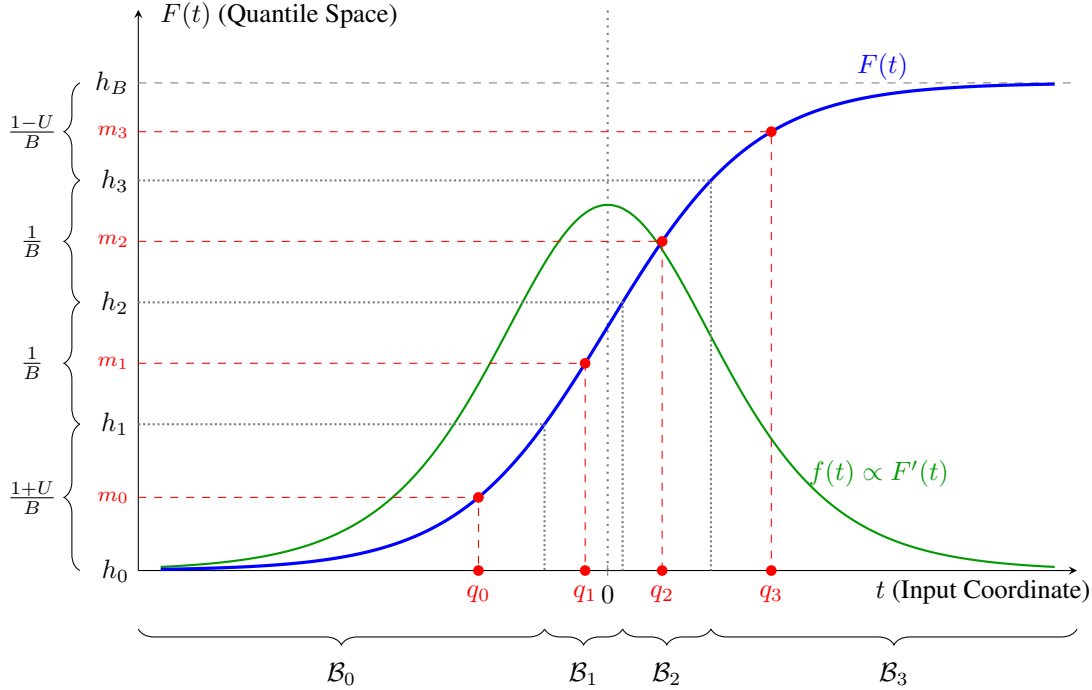


Figure 1: Illustration of the scalar quantizer mapping with $B = 4$. The function $F(t)$ (blue) and its scaled underlying density $f(t)$ (green) are shown. A uniform grid shifted by U defines boundaries h_j on the y-axis, which project back to the boundaries of the quantization buckets $\mathcal{B}_j$. The reconstruction values q_j are defined as the inverse image of the bucket midpoints $m_j = (h_j + h_{j+1})/2$.

Our vector quantization algorithm VECTORQUANT, presented in Algorithm 1, applies the scalar dithered quantization scheme described above to every coordinate of the input vector $x \in \mathcal{S}^{d-1}$ after applying the HD matrix.

For the purposes of the analysis it is convenient to think of the quantizer as returning the corresponding middle quantile as opposed to its index. To this effect, it is convenient to introduce

Definition 2.1 (quant function). For $z \in \mathbb{R}$ define $\text{quant}(z)$ to return the middle quantile q_j of the bucket $\mathcal{B}_j$ that contains z , i.e. $\text{quant}(z) := q_j$ for $z \in \mathcal{B}_j$.

With Definition 2.1 in place the task of bounding quantization error in all coordinates becomes, letting $z = HDx$, the task of bounding

$$\mathbb{E}_{\varepsilon, U} \sum_{i \in [d]} (z_i - \text{quant}(z_i))^2 = d \cdot \mathbb{E}_{\varepsilon, U} (z_1 - \text{quant}(z_1))^2,$$

where the last equality uses the fact that the marginal distributions of coordinates of z are the same, i.e. they are all distributed as the Rademacher sum $\sum_{j \in [d]} \varepsilon_j x_j$.

The details of the algorithms are depicted in Algorithm 1. Recall $[B]$ always denotes $\{0, 1, \dots, B-1\}$.

Algorithm 1 Dithered scalar Gaussian quantization

```

1: input: dimension  $d$  and bit-width  $b$ 
2: Let  $H \in \mathbb{R}^{d \times d}$  be a normalized Hadamard matrix and let  $B \leftarrow 2^b$ .
3: Sample independent Rademacher random variables  $\varepsilon_1, \dots, \varepsilon_d$  and set  $D \leftarrow \text{diag}(\varepsilon_1, \dots, \varepsilon_d)$ .
4: Sample an offset  $U \sim \text{Unif}[0, 1)$ .

5: procedure CONSTRUCTSCALARCODEBOOK( $B, U$ )
6:   Define grid points  $h_j \leftarrow (j + U)/B$  for  $j \in [B]$ . ▷ Uniformly shifted grid in  $[0, 1]$ 
   Define  $h_0 \leftarrow 0$  and  $h_B \leftarrow 1$ .
7:   Define buckets  $\mathcal{B}_j \leftarrow \{t \in \mathbb{R} : h_j \leq F(t) < h_{j+1}\}$  for  $j \in [B]$ .
8:   Define reconstruction values  $q_j \leftarrow F^{-1}((h_j + h_{j+1})/2)$  for  $j \in [B]$ .

9: procedure VECTORQUANT( $x \in S^{d-1}$ )
10:   $y \leftarrow HDx$ 
11:  for  $i \in [d]$  do
12:     $z_i \leftarrow \sqrt{d} y_i$ 
13:     $\text{id}_{x_i} \leftarrow \text{index } j \in [B] \text{ such that } z_i \in \mathcal{B}_j \text{ (i.e., the index of } z_i \text{'s bucket)}$ 
14:  return  $\text{id}_x = (\text{id}_{x_1}, \dots, \text{id}_{x_d}) \in [B]^d$ 

15: procedure VECTORDEQUANT( $\text{id}_x$ )
16:  for  $i \in [d]$  do
17:     $\tilde{y}_i \leftarrow (1/\sqrt{d}) \cdot q_{\text{id}_{x_i}}$  ▷  $q_{\text{id}_{x_i}}$  is the middle quantile of bucket  $\mathcal{B}_{\text{id}_{x_i}}$ , see Eq. (2)
18:   $\tilde{x} \leftarrow DH^\top \tilde{y}$ 
19:  return  $\tilde{x}$ 

```

The main guarantee of the Algorithm 1 is the following Theorem 2.2.

Theorem 2.2. For every dimension d and every unit vector $x \in S^{d-1}$, Algorithm 1 satisfies

$$\mathbb{E}_{D, U} \|x - \tilde{x}\|_2^2 \leq \frac{\pi\sqrt{3}/2 + o(1)}{4^b}, \quad b \rightarrow \infty,$$

where the $o(1)$ term vanishes as $b \rightarrow \infty$, uniformly over d and x .

To achieve the additional unbiasedness property, Algorithm 5 makes two changes to the scalar quantizer. First, instead of pinning the grid endpoints h_0, h_B at 0 and 1, it uses a shifted grid directly on the quantile levels $F(t)$, so that every input t has a uniformly random position within a bucket. Second, instead of reconstructing each bucket at the corresponding middle quantile by applying F^{-1} to the midpoint, it uses a modified reconstruction map G , carefully chosen so that averaging G over one bucket recovers the middle quantile value. This makes the scalar reconstruction unbiased, and hence the vector estimator unbiased, while G remains close enough to F^{-1} on the central region to preserve the same MSE bound. For details, see Algorithm 5 in Section A.3.

3 Proof sketch of Theorem 2.2

In this section, we outline our proof ideas for Theorem 2.2. We can use orthogonality of HD to decompose the error coordinate-wise:

$$\begin{aligned} \|x - \tilde{x}\|_2^2 &= \|x - DH^\top \tilde{y}\|_2^2 = \|HDx - \tilde{y}\|_2^2 = \\ &= \sum_{i=1}^d \left(y_i - \frac{1}{\sqrt{d}} \text{quant}(\sqrt{d}y_i) \right)^2 = \frac{1}{d} \sum_{i=1}^d \left(z_i - \text{quant}(z_i) \right)^2. \end{aligned}$$

Therefore, to control the total MSE: $\mathbb{E}_{D,U} \|x - \tilde{x}\|_2^2$, it is enough to bound the variance of the scalar quantizer $\text{quant}(\cdot)$ under the specific input distribution of z_i . For every coordinate i , this distribution is given by

$$z_i = \sqrt{d}(HDx)_i = \sqrt{d} \sum_j H_{ij} D_{jj} x_j = \sum_j \varepsilon_j x_j$$

where ε_j are i.i.d. Rademacher random variables taking values ± 1 with probability $1/2$. In contrast to using a truly random rotation or an i.i.d. Gaussian matrix, where z_i distributed as $\mathcal{N}(0, \|x\|_2^2)$, our case is more delicate, as distribution of $\sum_j \varepsilon_j x_j$ does not depend only on $\|x\|_2$, but also on individual coordinates. It means that our oblivious scalar quantizer quant should provide guarantees for *any* input distribution of the form $z_i = \sum_j \varepsilon_j x_j$, where $\|x\|_2 = 1$. We therefore drop index i in what follows. Recall that $B = 2^b$ is the number of buckets. Our main guarantee for quant is the following.

Lemma 3.1 (Restated Lemma A.1). *For every dimension d , every unit vector $x \in \mathcal{S}^{d-1}$ define $z = \sum_j \varepsilon_j x_j$ where i.i.d. $\varepsilon_j = \pm 1$ with probability $1/2$. Then*

$$\mathbb{E}_{\varepsilon,U} |z - \text{quant}(z)|^2 \leq \frac{\pi\sqrt{3}/2 + o(1)}{B^2}, \quad B \rightarrow \infty,$$

where the $o(1)$ term is uniform in d and x .

As we described above, Theorem 1.1 follows directly from Lemma 3.1. The rest of this section is dedicated to Lemma 3.1.

Lemma 3.1 is a bit surprising - the codebook of quant is built from the optimal quantization nodes for $\mathcal{N}(0, 1)$, but we prove that the same error rate is achieved for a different distribution: the Rademacher sums $z = \sum_j \varepsilon_j x_j$. This distribution can be quite different from $\mathcal{N}(0, \|x\|_2^2)$ for worst-case x . For example if $x = (1, 0, \dots, 0)$ then z only takes values ± 1 . However, it turns out that to prove Lemma 3.1, we only need the fact that z is 1-subgaussian. This is captured by the following claim.

Lemma 3.2. *For every dimension d , every unit vector $x \in \mathcal{S}^{d-1}$ define $z = \sum_j \varepsilon_j x_j$ where i.i.d. $\varepsilon_j = \pm 1$ with probability $1/2$. Then z is a 1-subgaussian random variable, i.e.*

$$\mathbb{E} e^{\lambda z} \leq e^{\lambda^2/2} \quad \text{for all } \lambda \in \mathbb{R}.$$

Note that subgaussianity is a powerful property that in some cases allows us to simply substitute the distribution with the standard Gaussian. More formally, we will use the following (see Lemma A.2):

Claim 3.3. *If $\alpha > 0$, z is 1-subgaussian and $\xi \sim \mathcal{N}(0, 1)$, then $\mathbb{E} e^{\alpha z^2} \leq \mathbb{E} e^{\alpha \xi^2}$.*

Unfortunately, there is no straightforward way to apply this property to bound the expectation of the quantization error because the error function $\min_{q_j} |t - q_j|^2$ is highly non-monotone – it could be that subgaussian distribution tends to concentrate around highs of the error function.

We now move on to outline our proof for Lemma 3.1. In fact, we will prove a bit more general version of Lemma 3.1 with any 1-subgaussian input.² From now on, we will write $\mathbb{E}_z$ instead of $\mathbb{E}_\varepsilon$.

We separate the analysis of $\mathbb{E}_{z,U} |z - \text{quant}(z)|^2$ according to whether z lies in a “central interval” $[-M, M]$, i.e. we separately consider a *tail* event $|z| > M$ and a *central* event $|z| \leq M$, for some carefully chosen threshold $M = C\sqrt{\log B}$, $C \in (4, 6)$. Such a narrow choice of values of C is due to the following tension: we need to guarantee that a $O(\frac{1}{B})$ shift in quantiles of F (see Algorithm 1) results in a $o(1)$ shift in centroids in the central interval; this forces M not to be too large. But with the same choice of M we need the tail event to be unlikely enough that we can ignore the centroids construction outside of the central interval completely; this forces M to be not too small.

²Note that $\mathcal{N}(0, 1)$ appearing from a truly random rotation is also 1-subgaussian.

3.1 Central event

In the *central* event, we argue that buckets are narrow enough to treat standard Gaussian density on them as uniform. Then, to recover optimal error rate for our arbitrary 1-subgaussian distribution, we induce uniformity by *dithering*, i.e. applying random shift on the quantization grid. More specifically, as we show in Lemma A.3, for our choice of M , any $|t| < M$ and quantile shift $\frac{U}{B} \sim \text{Unif}[0; 1/B]$ we have the following

- $T := \left[F(t) - F(\text{quant}(t)) \right]$ is uniform on $[-\frac{1}{2B}; \frac{1}{2B}]$.
- $\frac{F'(s)}{F'(t)} = 1 + o_B(1)$ for all $s \in [t; \text{quant}(t)]$ where $o_B(1)$ term is uniform in s and U .

Then we can use first-order approximation and write

$$\text{quant}(t) - t = F^{-1} \left(F(t) + \frac{T}{B} \right) - t = (1 + o_B(1)) \frac{T/B}{F'(t)}.$$

And, integrating over T , we bound

$$\mathbb{E}_U |t - \text{quant}(t)|^2 \leq (1 + o_B(1)) \frac{1}{12B^2} \frac{1}{F'(t)^2}.$$

Note that the error $\frac{T/B}{F'(t)}$, $T \sim \text{Unif}[-\frac{1}{2}, \frac{1}{2}]$ matches the typical error in a bucket of width $\frac{1}{BF'(t)}$ with a centroid placed in the middle. Since $F'(s)$ is almost constant over the bucket, this calculation differs from an actual error only by lower order terms.

To get the total error bound we first take expectation over U

$$\mathbb{E}_{z,U} |z - \text{quant}(z)|^2 \leq (1 + o_B(1)) \frac{1}{12B^2} \mathbb{E}_z \frac{1}{F'(z)^2}$$

and then, by Claim 3.3, since $F'(x)$ is a Gaussian density and z is 1-subgaussian,

$$\mathbb{E}_z \frac{1}{F'(z)^2} \leq \mathbb{E}_\xi \frac{1}{F'(\xi)^2}, \quad \text{where } \xi \sim \mathcal{N}(0, 1).$$

Finally,

$$\mathbb{E}_{z,U} |z - \text{quant}(z)|^2 \leq (1 + o_B(1)) \frac{1}{12B^2} \mathbb{E}_\xi \frac{1}{F'(\xi)^2} = \frac{\pi\sqrt{3}/2 + o(1)}{B^2}$$

3.2 Tail event

In the *tail* event, we give a simple bound

$$|z - \text{quant}(z)|^2 \leq 2|z|^2 + 2|\text{quant}(z)|^2$$

and argue that both terms on the RHS are bounded in expectation by $O(\text{poly log } B \cdot e^{-M^2/2}) = o(\frac{1}{B^2})$ using subgaussianity of z and our choice of M . A little more work is required to carefully bound $|\text{quant}(z)|$ since the centroid construction is randomized. For more details, see Lemma A.6.

4 Inner product quantizer

In this section we present our inner product quantizer and prove Theorem 1.2. For a vector $y \in \mathbb{R}^d$, in order to achieve approximation error for $(\langle y, x \rangle - \langle y, \tilde{x} \rangle)^2$ that scales as $\frac{O(1)}{dB^2} \|y\|_2^2$, we augment our quantizer for x with another basic quantizer that decorrelates the residual $r = x - \tilde{x}$ with y . This is achieved by a single application of a fresh HD matrix to the residual followed by a simple quantization procedure that uses a constant number of bits per coordinate in expectation. To obtain an explicit bound on the norm of the residual $\|r\|_2$, $\tilde{x}$ is projected onto the unit ball; such a projection $\Pi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ can be implemented simply as $\Pi(v) = v$ if $\|v\|_2 \leq 1$ and $\Pi(v) = v/\|v\|_2$ otherwise.

Concretely, let $r = x - \tilde{x}$ be the residual vector, where $\tilde{x}$ is the result of quantization using our main VECTORQUANT procedure followed by projecting onto the unit ball. Define the *residual scale* by

$s = \frac{\|r\|}{\sqrt{d}} \in [0, \frac{2}{\sqrt{d}}]$. We first quantize the scale using a basic scalar quantizer $\text{SCALARQUANT}(s)$ (Algorithm 2 below) and then quantize the residual's coordinates based on the quantized scale (Algorithm 3 below).

Algorithm 2 Scalar quantization

```

1: input: scalar  $s \in [0, \frac{2}{\sqrt{d}}]$ , dimension  $d$ , scale parameter  $B \geq 2$ 
2: public parameter: minimum encoded scale  $\tau_B \leftarrow 1/(dB)$ .

3: procedure SCALARQUANT( $s$ )
4:   if  $s < \tau_B$  then
5:     return  $\text{idx}_\sigma \leftarrow 0$ 
6:    $k \leftarrow \lceil \log_2(s/\tau_B) \rceil$ 
7:    $\text{idx}_\sigma \leftarrow k + 1$ 
8:   return  $\text{idx}_\sigma$ 

9: procedure SCALARDEQUANT( $\text{idx}_\sigma$ )
10:  if  $\text{idx}_\sigma = 0$  then
11:    return  $\sigma \leftarrow 0$ 
12:  return  $\sigma \leftarrow \tau_B 2^{\text{idx}_\sigma - 1}$ 

```

The intuition behind Algorithm 2 is that if the residual scale exceeds a minimum cutoff, i.e., $s \geq \tau_B$, then the quantized scale is $\sigma = \tau_B 2^{\text{idx}_\sigma - 1}$ satisfying $s \leq \sigma < 2s$. On the other hand if $s < \tau_B$, then the quantized scale is set to $\sigma = 0$ indicating that the residual is negligibly small (thus the residual quantization stage can be skipped). The quantized scales are indexed by idx_σ , which can be encoded using $\lceil \log_2(\lceil \log_2(\frac{2}{\sqrt{d}}/\tau_B) \rceil + 2) \rceil = O(\log \log(dB))$ bits when $s \in [0, \frac{2}{\sqrt{d}}]$.

We next describe the residual quantizer in Algorithm 3, which uses Algorithm 2 as a subroutine.

Algorithm 3 Residual Quantization

```

1: input: dimension  $d$ , scale parameter  $B \geq 2$ 
2: Let  $H \in \mathbb{R}^{d \times d}$  be a normalized Hadamard matrix and  $D$  be a diagonal matrix whose entries are i.i.d. Rademacher random variables.

3: procedure RESIDUALQUANT( $r \in \mathbb{R}^d$ )
4:    $\text{idx}_\sigma \leftarrow \text{SCALARQUANT}(\|r\|/\sqrt{d})$ 
5:   if  $\text{idx}_\sigma = 0$  then
6:     return  $\text{idx}_\sigma, \ell = (0, \dots, 0)$ , and  $\lambda = (0, \dots, 0)$ 
7:    $\sigma \leftarrow \text{SCALARDEQUANT}(\text{idx}_\sigma)$ 
8:    $v \leftarrow HD r$ 
9:   for  $i \in [d]$  do
10:     $\ell_i \leftarrow \min\{\ell \in \{0, 1, 2, \dots\} : |v_i| \leq \sigma \cdot 2^\ell\}$   $\triangleright \ell_i$  can be stored, e.g., as unary word  $1^{\ell_i}0$ 
11:     $R_i \leftarrow \sigma \cdot 2^{\ell_i}$ 
12:    Sample a sign bit  $\lambda_i \in \{-1, 1\}$  such that  $\mathbb{P}(\lambda_i = 1 \mid D, r) = \frac{1+(v_i/R_i)}{2}$ 
13:  return  $\text{idx}_\sigma, \ell = (\ell_1, \dots, \ell_d)$ , and  $\lambda = (\lambda_1, \dots, \lambda_d)$ 

14: procedure RESIDUALDEQUANT( $\text{idx}_\sigma, \ell, \lambda$ )
15:   $\sigma \leftarrow \text{SCALARDEQUANT}(\text{idx}_\sigma)$ 
16:  for  $i \in [d]$  do
17:     $R_i \leftarrow \sigma \cdot 2^{\ell_i}$ 
18:     $q_i \leftarrow R_i \lambda_i$ 
19:   $\hat{r} \leftarrow DH^\top q$ 
20:  return  $\hat{r}$ 

```

In English, for each coordinate $i \in [d]$, Algorithm 3 records the level ℓ_i , for instance using the unary word $1^{\ell_i}0$, and sample one randomized sign bit $\lambda_i \in \{-1, 1\}$ with $\mathbb{P}(\lambda_i = 1 \mid D, r) = \frac{1+(v_i/R_i)}{2}$. This probability is valid because $|v_i| \leq \sigma 2^{\ell_i} = R_i$.

Finally, our two-stage inner product quantizer is present below as Algorithm 4. Its analysis, namely the proof of Theorem 1.2, is presented in the appendix (Section B).

Algorithm 4 Two-Stage Inner Product Quantizer

```

1: input: dimension  $d$ , base bit-width  $b$ 
2: Let  $H \in \mathbb{R}^{d \times d}$  be a normalized Hadamard matrix, let  $B \leftarrow 2^b$ , and let  $\Pi : \mathbb{R}^d \rightarrow \mathbb{R}^d$  be a projection to the unit ball.
3: Sample independent Rademacher random variables  $\varepsilon_1, \dots, \varepsilon_d$  and set  $D_{\text{base}} \leftarrow \text{diag}(\varepsilon_1, \dots, \varepsilon_d)$ .
4: Sample independent Rademacher random variables  $\eta_1, \dots, \eta_d$  and set  $D_{\text{res}} \leftarrow \text{diag}(\eta_1, \dots, \eta_d)$ .
5: Sample an offset  $U \sim \text{Unif}[0, 1)$ .
6: CONSTRUCTSCALARCODEBOOK( $B, U$ ) ▷ Defined in Algorithm 1. This constructs  $\mathcal{B}_j$  and  $q_j$  for  $j \in [B]$ 

```

```

7: procedure QUANTIZE( $x \in \mathcal{S}^{d-1}$ )
8: ▷ Stage 1: Base Dithered Quantization
9:    $y \leftarrow HD_{\text{base}}x$ 
10:  for  $i \in [d]$  do
11:     $z_i \leftarrow \sqrt{d} y_i$ 
12:     $\text{idx}_i \leftarrow \text{index } j \in [B] \text{ such that } z_i \in \mathcal{B}_j$ 
13:     $\tilde{y}_i \leftarrow (1/\sqrt{d}) \cdot q_{\text{idx}_i}$  ▷ Local dequantization for residual
14:   $\text{idx} \leftarrow (\text{idx}_1, \dots, \text{idx}_d) \in [B]^d$ 
15:   $\tilde{x} \leftarrow \Pi D_{\text{base}} H^\top \tilde{y}$  ▷ Base reconstructed vector
16: ▷ Stage 2: Residual Quantization, repeated from Algorithm 3
17:   $r \leftarrow x - \tilde{x}$  ▷ Compute the residual vector
18:   $\text{idx}_\sigma \leftarrow \text{SCALARQUANT}(\|r\|/\sqrt{d})$ 
19:  if  $\text{idx}_\sigma = 0$  then
20:    return  $\text{idx}, \text{idx}_\sigma, \ell = (0, \dots, 0), \lambda = (0, \dots, 0)$ 
21:   $\sigma \leftarrow \text{SCALARDEQUANT}(\text{idx}_\sigma)$ 
22:   $v \leftarrow HD_{\text{res}}r$ 
23:  for  $i \in [d]$  do
24:     $\ell_i \leftarrow \min\{\ell \in \{0, 1, 2, \dots\} : |v_i| \leq \sigma 2^\ell\}$  ▷  $\ell_i$  can be stored, e.g., as unary word  $1^{\ell_i}0$ 
25:     $R_i \leftarrow \sigma \cdot 2^{\ell_i}$ 
26:    Sample sign bit  $\lambda_i \in \{-1, 1\}$  such that  $\mathbb{P}(\lambda_i = 1 \mid D_{\text{res}}, r) = \frac{1+v_i/R_i}{2}$ 
27:  return  $\text{idx}, \text{idx}_\sigma, \ell = (\ell_1, \dots, \ell_d)$ , and  $\lambda = (\lambda_1, \dots, \lambda_d)$ 

```

```

28: procedure DEQUANTIZE( $\text{idx}, \text{idx}_\sigma, \ell, \lambda$ )
29: ▷ Stage 1: Base Reconstruction
30:  for  $i \in [d]$  do
31:     $\tilde{y}_i \leftarrow (1/\sqrt{d}) \cdot q_{\text{idx}_i}$ 
32:   $\tilde{x} \leftarrow \Pi D_{\text{base}} H^\top \tilde{y}$ 
33: ▷ Stage 2: Residual Reconstruction, repeated from Algorithm 3
34:   $\sigma \leftarrow \text{SCALARDEQUANT}(\text{idx}_\sigma)$ 
35:  if  $\sigma = 0$  then
36:     $\hat{r} \leftarrow 0$ 
37:  else
38:    for  $i \in [d]$  do
39:       $R_i \leftarrow \sigma 2^{\ell_i}$ 
40:       $q_i^{\text{res}} \leftarrow R_i \lambda_i$ 
41:     $\hat{r} \leftarrow D_{\text{res}} H^\top q^{\text{res}}$ 
42:  return  $\hat{x} \leftarrow \tilde{x} + \hat{r}$  ▷ Final reconstructed vector

```

Limitations

This paper is primarily theoretical and studies a specific class of data-oblivious quantizers based on a single randomized Hadamard transform and randomized dithering. The results characterize worst-case mean-squared error and inner-product error, but do not attempt to optimize all finite-bit constants or evaluate performance empirically across downstream systems. Extending the analysis to broader structured transforms, alternative scalar quantizers, and more implementation-specific bit accounting would be interesting directions for future work.

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A Full proof of Theorem 1.1

A.1 Scalar Estimate

In this section, we prove Lemma A.1, the scalar estimate that controls the coordinate-wise mean squared error. In the next section, we then use it to bound the error of vector quantization.

For any unit vector $a \in \mathbb{R}^d$, we define a random variable

$$X_a := \sum_{i=1}^d a_i \varepsilon_i,$$

where $\varepsilon_1, \dots, \varepsilon_d$ are independent Rademacher random variables. We abbreviate $\varepsilon = (\varepsilon_1, \dots, \varepsilon_d)$. For the scalar quantizer appearing in Algorithm 1, we write

$$\text{quant}(t) := q_j \quad \text{when } t \in \mathcal{B}_j,$$

i.e. $\text{quant}(\cdot)$ maps any scalar $t \in \mathbb{R}$ to the corresponding reconstruction value, which is the preimage of the midpoint of the bucket in which $F(t)$ lands. We recall that quant depends on U, B in the following ways: U is a random offset that determines the shift of buckets, and $B = 2^b$ is the number of buckets. The rest of this section devotes to proving Lemma A.1, which bounds the mean squared quantization error for quant applied to X_a .

Lemma A.1. *Uniformly over all $d \geq 1$ and all $a \in \mathbb{R}^d$ satisfying $\sum_i a_i^2 = 1$,*

$$\mathbb{E}_{\varepsilon, U} (X_a - \text{quant}(X_a))^2 \leq \frac{\pi\sqrt{3}/2 + o(1)}{B^2}, \quad B \rightarrow \infty.$$

As outlined in the proof sketch, the proof proceeds by separating the behavior of the scalar quantizer according to whether the input lies in a “central” interval $[-M_B, M_B]$, where $M_B := \sqrt{5 \log B}$. We first show in Lemma A.2 that every normalized Rademacher sum has Gaussian-type tail and moment bounds. Inside the central interval, Lemma A.3 shows that the random shift U makes the displacement in the F -coordinate uniform at scale $1/B$, and that F^{-1} can be replaced by its first-order approximation on this scale. Outside the central interval, Lemma A.6 gives a rough bound on the squared quantization error; the tail bound from Lemma A.2 then makes this rough estimate negligible. Combining these two regimes gives Lemma A.1.

The following lemma is standard (see e.g., [AC09], proof of Lemma 1), but we reprove it here for completeness.

Lemma A.2. *For every unit vector $a \in \mathbb{R}^d$, the random variable X_a satisfies*

$$\mathbb{E} e^{\lambda X_a} \leq e^{\lambda^2/2} \quad \text{for all } \lambda \in \mathbb{R}.$$

Consequently,

$$\mathbb{P}(|X_a| > s) \leq 2e^{-s^2/2} \quad \text{for all } s > 0.$$

Moreover, if $G \sim \mathcal{N}(0, 1)$, then

$$\mathbb{E} e^{X_a^2/3} \leq \mathbb{E} e^{G^2/3} = \sqrt{3}.$$

Proof. By the definition of X_a and the independence between ε_i ’s,

$$\mathbb{E} e^{\lambda X_a} = \mathbb{E} e^{\lambda \sum_{i=1}^d a_i \varepsilon_i} = \prod_{i=1}^d \mathbb{E} e^{\lambda a_i \varepsilon_i}.$$

For each Rademacher random variable ε_i ,

$$\mathbb{E} e^{\lambda a_i \varepsilon_i} = \frac{1}{2} e^{\lambda a_i} + \frac{1}{2} e^{-\lambda a_i} = \cosh(\lambda a_i) \leq e^{(\lambda a_i)^2/2}.$$

Thus

$$\mathbb{E} e^{\lambda X_a} \leq \prod_{i=1}^d e^{(\lambda a_i)^2/2} = e^{\sum_{i=1}^d a_i^2 \lambda^2/2} = e^{\lambda^2/2}. \quad (3)$$

The tail bound follows from Markov's inequality applied to both $e^{\lambda X_a}$ and $e^{-\lambda X_a}$ and optimize over $\lambda > 0$.

For the last argument, since $\mathbb{E}_{G \sim \mathcal{N}(0,1)} e^{\tau G} = e^{\tau^2/2}$ for any $\tau \in \mathbb{R}$, conditioning on X_a and taking $\tau = \sqrt{2/3} X_a$ gives

$$e^{X_a^2/3} = \mathbb{E}_G e^{\sqrt{2/3} G X_a}.$$

Let G be independent of $\varepsilon_1, \dots, \varepsilon_d$, we obtain

$$\mathbb{E} e^{X_a^2/3} = \mathbb{E}_G \mathbb{E}_\varepsilon e^{\sqrt{2/3} G X_a} \leq \mathbb{E}_G e^{G^2/3} = \sqrt{3}.$$

where the inequality follows from the earlier MGF bound Eq. (3). $\square$

We next state our main lemma for the central event. It controls the quantization error for all real t (in particular, our random variable X_a) that lies in the central interval $[-M_B, M_B]$.

Lemma A.3. *Let $M_B := \sqrt{5 \log B}$. For all $|t| \leq M_B$,*

$$\mathbb{E}_U (t - \text{quant}(t))^2 \leq (1 + o_B(1)) \frac{A^2}{12B^2} (2\pi)^{1/3} e^{t^2/3}.$$

where the $o_B(1)$ term is uniform for $|t| \leq M_B$.

Before proving Lemma A.3, we state the following fact about F , which will be helpful for our proof. Let Φ denote the CDF of the standard normal distribution.

Claim A.4. *For any $t \in \mathbb{R}$, $F(t) = \Phi(t/\sqrt{3})$.*

Proof. Recall that

$$F(t) := \frac{1}{A} \int_{-\infty}^t \varphi(s)^{1/3} ds.$$

where $A = \int_{\mathbb{R}} \varphi(t)^{1/3} dt$ is the normalization factor. Since

$$\varphi(s)^{1/3} = (2\pi)^{-1/6} e^{-s^2/6}$$

is proportional to the density of $\mathcal{N}(0, 3)$, $F(t)$ is exactly the CDF of $\mathcal{N}(0, 3)$. Thus for $G \sim \mathcal{N}(0, 1)$,

$$F(t) = \mathbb{P}[\sqrt{3}G \leq t] = \mathbb{P}[G \leq t/\sqrt{3}] = \Phi(t/\sqrt{3}).$$

$\square$

Corollary A.5. *Let $M_B = \sqrt{5 \log B}$. We have $F(-M_B)$, $1 - F(M_B)$, $F(M_B + 1) - F(M_B)$, and $F(-M_B) - F(-M_B - 1) \gg B^{-1}$.*

Proof. By Claim A.4 and the symmetry of the Gaussian distribution,

$$F(-M_B) = 1 - F(M_B) = \Phi(-M_B/\sqrt{3}).$$

Since $M_B^2 = 5 \log B$, the Gaussian tail estimate gives

$$\Phi(-M_B/\sqrt{3}) = B^{-5/6+o(1)}.$$

which is in particular $\omega(B^{-1})$. For the two increments,

$$\begin{aligned} F(M_B + 1) - F(M_B) &= \int_{M_B}^{M_B+1} F'(t) dt \\ &= \int_{M_B}^{M_B+1} \frac{(2\pi)^{-1/6}}{A} \exp(-t^2/6) dt \\ &\geq \frac{(2\pi)^{-1/6}}{A} \exp(-(M_B + 1)^2/6) \end{aligned}$$

where the second equality follows from the definition of F . Since $M_B^2 = 5 \log B$ and A is a constant, this is $B^{-5/6-o(1)} = \omega(B^{-1})$ as claimed. By the symmetry, the same bound holds for $F(-M_B) - F(-M_B - 1)$. $\square$

Proof of Lemma A.3. Recall that when defining the grid points in Algorithm 1, we set $h_1 = (1 + U)/B$ and $h_{B-1} = (B - 1 + U)/B$. As U varies in $[0, 1)$, we always have $h_1 \leq 2/B$ and $h_{B-1} \geq 1 - (1/B)$. On the other hand, since F is increasing, for $|t| \leq M_B$, we have $F(-M_B) \leq F(t) \leq F(M_B)$. Applying Corollary A.5, we get

$$\omega(B^{-1}) \leq F(t) \leq 1 - \omega(B^{-1}).$$

So for all sufficiently large B , $h_1 < F(t) < h_{B-1}$, i.e. $F(t)$ never falls in one of the two endpoint buckets.

Therefore the effect of the random offset U is to uniformly choose the relative position of $F(t)$ inside a bucket. Equivalently, the midpoint of the bucket containing $F(t)$ can be written as a random variable

$$F(t) + \frac{T}{B},$$

where T is uniformly distributed on $[-1/2, 1/2]$. And its preimage is

$$\text{quant}(t) = F^{-1} \left(F(t) + \frac{T}{B} \right).$$

We next estimate $\text{quant}(t) - t$ using the first-order Taylor expansion of F^{-1} at the point $F(t)$. Formally, we claim that, uniformly over $|t| \leq M_B$ and $|T/B| \leq 1/(2B)$,

$$\text{quant}(t) - t = F^{-1} \left(F(t) + \frac{T}{B} \right) - t = (1 + o_B(1)) \frac{T/B}{F'(t)}.$$

To show this, we observe that for $s \in [0, T/B]$, the point $F^{-1}(F(t) + s)$ lies in the interval $[-M_B - 1, M_B + 1]$, for all sufficiently large B . Indeed, $F(M_B + 1) - F(M_B)$ and $F(-M_B) - F(-M_B - 1) \gg B^{-1}$ by Corollary A.5. So for $|s| \leq |T/B| \leq 1/(2B)$, moving by s in the F -coordinate cannot move the preimage past $M_B + 1$ or $-M_B - 1$.

Now let $y = F^{-1}(F(t) + s)$. The mean value theorem gives

$$s = F'(\xi)(y - t)$$

for some ξ between y and t . By the previous argument, both endpoints lie in $[-M_B - 1, M_B + 1]$, and hence so does ξ . Therefore,

$$|y - t| \leq |s| \sup_{|\xi| \leq M_B + 1} \frac{1}{F'(\xi)} \leq \Theta \left(B^{-1} \exp \left(\frac{(M_B + 1)^2}{6} \right) \right) = B^{-1/6+o(1)}$$

where the second step follows from $F'(\xi) = \frac{(2\pi)^{-1/6}}{A} e^{-\xi^2/6}$. This implies

$$|y^2 - t^2| = |y - t| |y + t| \leq B^{-1/6+o(1)} \cdot O(M_B) = B^{-1/6+o(1)} \cdot O(\sqrt{\log B}) = o_B(1)$$

thus

$$\frac{F'(t)}{F'(y)} = \frac{e^{-t^2/6}}{e^{-y^2/6}} = \exp \left(\frac{y^2 - t^2}{6} \right) = 1 + o_B(1)$$

uniformly over $|t| \leq M_B$ and $|s| \leq 1/(2B)$. Equivalently,

$$\frac{1}{F'(F^{-1}(F(t) + s))} = (1 + o_B(1)) \frac{1}{F'(t)}.$$

Finally, integrating from 0 to T/B gives

$$F^{-1} \left(F(t) + \frac{T}{B} \right) - t = \int_0^{T/B} \frac{1}{F'(F^{-1}(F(t) + s))} ds = (1 + o_B(1)) \frac{T/B}{F'(t)}. \quad (4)$$

Using $\mathbb{E}T^2 = 1/12$ and $F'(t) = \frac{(2\pi)^{-1/6}}{A} e^{-t^2/6}$, we obtain

$$\mathbb{E}_U(t - \text{quant}(t))^2 \leq (1 + o_B(1)) \frac{1}{12B^2} \frac{1}{F'(t)^2} = (1 + o_B(1)) \frac{A^2}{12B^2} (2\pi)^{1/3} e^{t^2/3}.$$

□

Next we consider the tail event when $|t| = |X_a| > M_B$. In this event, we use the following crude error bound, which suffices for the entire proof due to the small probability of the tail event.

Lemma A.6. *There is an absolute constant C such that, for every $B \geq 2$ and every $t \in \mathbb{R}$,*

$$\mathbb{E}_U(t - \text{quant}(t))^2 \leq C(t^2 + \log B).$$

Proof. Fix the value of U . Since F^{-1} is increasing, the reconstruction values $q_0, \dots, q_{B-1}$ are increasing. Hence every reconstruction value lies between the two endpoint reconstruction values, and therefore

$$|\text{quant}(t)| \leq R_B(U),$$

where

$$R_B(U) := \max \left\{ \left| F^{-1} \left(\frac{1+U}{2B} \right) \right|, \left| F^{-1} \left(1 - \frac{1-U}{2B} \right) \right| \right\}.$$

By Claim A.4, we have $F^{-1}(p) = \sqrt{3} \Phi^{-1}(p)$. Using the standard Gaussian quantile bound

$$|\Phi^{-1}(r)|^2 + |\Phi^{-1}(1-r)|^2 \leq C \log(1/r), \quad 0 < r \leq 1/2,$$

we obtain

$$R_B(U)^2 \leq C \log \left(\frac{2B}{1+U} \right) + C \log \left(\frac{2B}{1-U} \right).$$

Taking expectation over U ,

$$\mathbb{E}_U R_B(U)^2 \leq C \int_0^1 \log(2B) \, dU + C \int_0^1 \log \left(\frac{2B}{1-U} \right) \, dU \leq C \log B.$$

Thus

$$\mathbb{E}_U(t - \text{quant}(t))^2 \leq 2t^2 + 2\mathbb{E}_U q(t)^2 \leq 2t^2 + 2\mathbb{E}_U R_B(U)^2 \leq C(t^2 + \log B).$$

□

With Lemma A.2, Lemma A.3, and Lemma A.6, we can now prove the scalar guarantee.

Proof of Lemma A.1. Fix $a \in \mathbb{R}^d$ with $\sum_i a_i^2 = 1$, and write $X = X_a$. We split the expectation according to the central event $|X| \leq M_B$.

We first consider the central contribution, i.e. when $|X| \leq M_B$. By Lemma A.3, and using the uniformity of the $o_B(1)$ term, there exists a deterministic sequence $\alpha_B \rightarrow 0$ such that

$$\begin{aligned} B^2 \mathbb{E}_{\varepsilon, U} [(X - \text{quant}(X))^2 \mathbf{1}_{\{|X| \leq M_B\}}] &\leq (1 + \alpha_B) \frac{A^2}{12} (2\pi)^{1/3} \mathbb{E}_{\varepsilon} e^{X^2/3} \\ &\leq (1 + \alpha_B) \frac{A^2}{12} (2\pi)^{1/3} \sqrt{3} \\ &= (1 + \alpha_B) \frac{A^3}{12} \end{aligned}$$

where the second inequality follows from $\mathbb{E}_{\varepsilon} e^{X^2/3} \leq \sqrt{3}$ as shown in Lemma A.2. Since $A = (2\pi)^{1/3} \sqrt{3}$, this is $(1 + \alpha_B) \frac{\pi \sqrt{3}}{2}$. It remains to show that the complement of the central event contributes $o(B^{-2})$. We use Lemma A.6:

$$B^2 \mathbb{E}_{\varepsilon, U} [(X - \text{quant}(X))^2 \mathbf{1}_{\{|X| > M_B\}}] \leq C B^2 \mathbb{E}_{\varepsilon} [(X^2 + \log B) \mathbf{1}_{\{|X| > M_B\}}].$$

Then using the tail bound from Lemma A.2,

$$\mathbb{E}_{\varepsilon} [X^2 \mathbf{1}_{\{|X| > M_B\}}] = M_B^2 \Pr[|X| > M_B] + \int_{M_B^2}^{\infty} \Pr[X^2 > v] \, dv \leq C(M_B^2 + 1)e^{-M_B^2/2}.$$

Also,

$$(\log B) \Pr[|X| > M_B] \leq C(\log B)e^{-M_B^2/2}.$$

Since $M_B^2 = 5 \log B$, this gives

$$B^2 \mathbb{E}_{\varepsilon, U} [(X - \text{quant}(X))^2 \mathbf{1}_{\{|X| > M_B\}}] \leq C B^2 (\log B) e^{-5 \log B/2} = C(\log B) B^{-1/2} = o(1).$$

Combining the central and tail contributions,

$$B^2 \mathbb{E}_{\varepsilon, U} (X_a - \text{quant}(X_a))^2 \leq \frac{\pi \sqrt{3}}{2} + o(1),$$

where the $o(1)$ term is uniform over d and a .

□

A.2 Proof of Theorem 2.2

We now return to the vector quantizer in Algorithm 1 and prove Theorem 2.2. The following lemma is the only property of the randomized Hadamard transform that we need: after the normalization, each coordinate is exactly a Rademacher sum with a unit coefficient vector.

Lemma A.7. Fix $x \in \mathcal{S}^{d-1}$, and let $y = HDx$ and $z_i = \sqrt{d}y_i$ for $i \in [d]$ as in Algorithm 1. For every $i \in [d]$, there exists a unit vector $a^{(i)} \in \mathbb{R}^d$ such that $z_i = X_{a^{(i)}}$.

Proof. By the definition of D , we have $D = \text{diag}(\varepsilon_1, \dots, \varepsilon_d)$. Thus

$$z_i = \sqrt{d}(HDx)_i = \sum_{j=1}^d \sqrt{d} H_{i,j} x_j \varepsilon_j.$$

Define

$$a_j^{(i)} := \sqrt{d} H_{i,j} x_j.$$

Since H is a normalized Hadamard matrix, $H_{i,j}^2 = 1/d$ for every i, j . Hence

$$\sum_{j=1}^d (a_j^{(i)})^2 = \sum_{j=1}^d d H_{i,j}^2 x_j^2 = \sum_{j=1}^d x_j^2 = 1.$$

Therefore $z_i = \sum_j a_j^{(i)} \varepsilon_j = X_{a^{(i)}}$, as claimed. $\square$

With this, we can directly apply our scalar estimate for each coordinate and obtain the mean squared error bound for the vector quantizer.

Proof of Theorem 1.1. Fix $x \in \mathcal{S}^{d-1}$. Let $y = HDx$, and let $z_i = \sqrt{d}y_i$ be the scalar input quantized in coordinate i . If $\tilde{y}$ denotes the vector reconstructed in the Hadamard domain, then, by the scaling in Algorithm 1,

$$\tilde{y}_i = \frac{1}{\sqrt{d}} \text{quant}(z_i).$$

Since H and D are orthogonal,

$$\|x - \tilde{x}\|_2^2 = \|HDx - \tilde{y}\|_2^2 = \frac{1}{d} \sum_{i=1}^d (z_i - \text{quant}(z_i))^2.$$

Taking expectation over D and U , and using linearity of expectation,

$$\mathbb{E}_{D,U} \|x - \tilde{x}\|_2^2 = \frac{1}{d} \sum_{i=1}^d \mathbb{E}_{D,U} (z_i - \text{quant}(z_i))^2.$$

By the Lemma A.7, each z_i is equal to $X_{a^{(i)}}$ for a unit vector $a^{(i)}$. Applying Lemma A.1 to each coordinate gives

$$\mathbb{E}_{D,U} (z_i - \text{quant}(z_i))^2 \leq \frac{\pi\sqrt{3}/2 + o(1)}{B^2}.$$

Then averaging over i , we obtain

$$\mathbb{E}_{D,U} \|x - \tilde{x}\|_2^2 \leq \frac{\pi\sqrt{3}/2 + o(1)}{B^2}.$$

$\square$

A.3 An unbiased variant of Algorithm 1

In this section we modify the scalar quantizer, in particular the scalar reconstruction rule, so that the resulting vector estimator is unbiased, while preserving the same asymptotic distortion bound. Recall that

$$F(t) = \frac{1}{A} \int_{-\infty}^t \phi(s)^{1/3} ds, \quad A = \int_{\mathbb{R}} \phi(s)^{1/3} ds.$$

Previously in Algorithm 1, we define buckets with $1/B$ spacing in the F -coordinates, and reconstruct each bucket at the inverse image (F^{-1}) of its midpoint. In this section, we will use buckets with spacing $1/(B-1)$, and we will replace F^{-1} with a reconstruction function $G : [-\frac{1}{2(B-1)}, 1 + \frac{1}{2(B-1)}] \rightarrow \mathbb{R}$ satisfying

$$(B-1) \int_{r - \frac{1}{2(B-1)}}^{r + \frac{1}{2(B-1)}} G(s) ds = F^{-1}(r) \quad \text{for every } r \in (0, 1). \quad (5)$$

Moreover, we will choose G such that in the central region $[-\sqrt{5 \log B}, \sqrt{5 \log B}]$, G is close to F^{-1} . Formally:

Lemma A.8. *There exists an explicitly defined function $G : [-\frac{1}{2(B-1)}, 1 + \frac{1}{2(B-1)}] \rightarrow \mathbb{R}$ that satisfies Eq. (5). Moreover, it can be chosen so that for all $|t| \leq \sqrt{5 \log B}$ and all $|v| \leq 1/2$,*

$$G\left(F(t) + \frac{v}{B-1}\right) = t + \frac{v}{(B-1)F'(t)} + o(1) \frac{1}{(B-1)F'(t)}. \quad (6)$$

Proof. We denote the spacing of the grid buckets $\delta := \frac{1}{B-1}$. We start by defining G on the open interval $I := (-\frac{\delta}{2}, 1 + \frac{\delta}{2})$ as follows: for $u \in (\frac{1-\delta}{2}, \frac{1+\delta}{2}]$ and $k \in \mathbb{Z}$, whenever $u + k\delta \in I$, we define G as

$$G(u + k \cdot \delta) := F^{-1}(u) + \begin{cases} \delta \sum_{j=0}^{k-1} (F^{-1})'(u + (j+1/2) \cdot \delta) & \text{if } k > 0 \\ 0 & \text{if } k = 0 \\ -\delta \sum_{j=k}^{-1} (F^{-1})'(u + (j+1/2) \cdot \delta) & \text{if } k < 0 \end{cases}.$$

The key idea behind this definition is to enforce the recurrence

$$G(u + (k+1)\delta) - G(u + k\delta) = \delta (F^{-1})'(u + (k+1/2)\delta).$$

We first argue that G satisfies Eq. (5). For every $r \in (0, 1)$, there is a unique pair of u and k such that we can write $r - \delta/2 = u + k \cdot \delta$, it follows that $G(r + \delta/2) - G(r - \delta/2) = \delta (F^{-1})'(r)$. Therefore,

$$\frac{d}{dr} \left(\frac{1}{\delta} \int_{r-\delta/2}^{r+\delta/2} G(s) ds \right) = \frac{1}{\delta} (G(r + \delta/2) - G(r - \delta/2)) = (F^{-1})'(r),$$

i.e.

$$C := \frac{1}{\delta} \int_{r-\delta/2}^{r+\delta/2} G(s) ds - F^{-1}(r)$$

is constant in r . Evaluating this at $r = 1/2$ shows that $C = 0$. Indeed, on $(\frac{1-\delta}{2}, \frac{1+\delta}{2}]$, $G = F^{-1}$, and $F(t) = \Phi(t/\sqrt{3})$ (Claim A.4) implies that $F^{-1}(1/2) = 0$ and $F^{-1}(1-r) = -F^{-1}(r)$ for every r .

This shows that G satisfies Eq. (5) on the open interval I . We extend it arbitrarily to the endpoints for notational convenience in the codebook. These endpoint values are irrelevant for Eq. (5) since they are measure-zero for the integral.

It remains to show Eq. (6). We first compare G with F^{-1} . Fix $r \in I$, and write $r = u + k\delta$ again with the unique pair of $u \in (\frac{1-\delta}{2}, \frac{1+\delta}{2}]$ and $k \in \mathbb{Z}$. In the following, we focus on $k \geq 0$, with the convention that $\sum_{j=0}^{-1}(\cdot) = 0$. The case $k < 0$ holds symmetrically. By the definition of G ,

$$G(r) - F^{-1}(r) = \sum_{j=0}^{k-1} \left[\delta (F^{-1})'(u + (j+1/2)\delta) - \int_{u+j\delta}^{u+(j+1)\delta} (F^{-1})'(s) ds \right].$$

Hence, by the midpoint rule error bound,

$$|G(r) - F^{-1}(r)| \leq O \left(\delta^3 \sum_{j=0}^{k-1} \sup_{s \in [u+j\delta, u+(j+1)\delta]} |(F^{-1})'''(s)| \right).$$

Equivalently, for J denoting the interval between u and r ,

$$|G(r) - F^{-1}(r)| \leq O \left(\delta^2 \int_J |(F^{-1})'''(s)| \, ds \right) + O \left(\delta^3 \sup_{s \in J} |(F^{-1})'''(s)| \right).$$

It remains to bound this RHS plus $F^{-1}(r) - t$, using our parameters from Eq. (6):

$$r = F(t) + v\delta, \quad |t| \leq M_B, \quad |v| \leq 1/2.$$

In Lemma A.3 (Eq. (4)), we have shown that $F^{-1}(r) - t = (1 + o(1)) \frac{v\delta}{F'(t)}$. Also, we have shown that $F^{-1}(r) \in [-M_B - 1, M_B + 1]$ for all sufficiently large B . Combining this with the fact that $F^{-1}(u) = O(\delta)$ for $u \in (\frac{1-\delta}{2}, \frac{1+\delta}{2}]$, we have that for every $s \in J$, $|F^{-1}(s)| \leq M_B + 1$. And by the definition of F , one can observe that

$$(F^{-1})'(s) = \frac{1}{F'(F^{-1}(s))}, \quad (F^{-1})''(s) = \frac{F^{-1}(s)}{3F'(F^{-1}(s))^2},$$

and

$$(F^{-1})'''(s) = \left(\frac{1}{3} + \frac{2(F^{-1}(s))^2}{9} \right) \frac{1}{F'(F^{-1}(s))^3}.$$

Combining all these, we get

$$\int_J |(F^{-1})'''(s)| \, ds \leq O \left(\frac{M_B}{F'(t)^2} \right), \quad \sup_{s \in J} |(F^{-1})'''(s)| \leq O \left(\frac{M_B^2}{F'(t)^3} \right).$$

Returning to Eq. (6), we get that

$$\begin{aligned} G(r) - t &\leq F^{-1}(r) - t + O \left(\delta^2 \int_J |(F^{-1})'''(s)| \, ds \right) + O \left(\delta^3 \sup_{s \in J} |(F^{-1})'''(s)| \right) \\ &\leq (1 + o(1)) \frac{v\delta}{F'(t)} + \delta^2 O \left(\frac{M_B}{F'(t)^2} \right) + \delta^3 O \left(\frac{M_B^2}{F'(t)^3} \right) \\ &= (1 + o(1)) \frac{v\delta}{F'(t)} + \text{poly log } B \cdot (B^{-1/6+o(1)} + B^{-1/3+o(1)}) \cdot \frac{\delta}{F'(t)} \\ &= \frac{v\delta}{F'(t)} + o(1) \frac{\delta}{F'(t)} \end{aligned}$$

where in the third step we used $M_B = \sqrt{5 \log B}$ and $\frac{\delta}{F'(t)} \leq \Theta(\delta \exp(M_B^2/6)) = B^{-1/6+o(1)}$. $\square$

We now describe the unbiased variant of Algorithm 1 in Algorithm 5. The only change is the procedure of CONSTRUCTSCALARCODEBOOK, so we omit repeating the procedures VECTORQUANT and VECTORDEQUANT.

Algorithm 5 Unbiased dithered scalar Gaussian quantization

- 1: **input:** dimension d and bit-width b
 - 2: Let $H \in \mathbb{R}^{d \times d}$ be a normalized Hadamard matrix and let $B \leftarrow 2^b$.
 - 3: Sample independent Rademacher random variables $\varepsilon_1, \dots, \varepsilon_d$ and set $D \leftarrow \text{diag}(\varepsilon_1, \dots, \varepsilon_d)$.
 - 4: Sample an offset $U \sim \text{Unif}[0, 1)$.
-

- 5: **procedure** CONSTRUCTSCALARCODEBOOK(B, U)
 - 6: Define buckets $\mathcal{B}_j \leftarrow \{t \in \mathbb{R} : \lfloor (B-1)F(t) - U \rfloor + 1 = j\}$ for $j = 0, \dots, B-1$.
 - 7: Define reconstruction values $q_j \leftarrow G(\frac{j+U-1/2}{B-1})$ for $j = 0, \dots, B-1$.
-

In English, we use a shifted grid in the F -coordinate with spacing $1/(B-1)$. For $r \in (0, 1)$, define grid points

$$h(r) = \lfloor (B-1)r - U \rfloor,$$

where $U \sim \text{Unif}[0, 1]$. We only need to focus on $r \in (0, 1)$ since later we will take $r = F(t)$, which is in $(0, 1)$ for any real t . Since $0 < r < 1$, the index satisfies $h(r) \in \{-1, 0, 1, \dots, B-2\}$. Thus $h(r) + 1 \in \{0, 1, \dots, B-1\}$, so the index is still encodable using $\log_2 B = b$ bits. We denote the midpoint of the shifted cell containing r

$$m(r) = \frac{h(r) + U + 1/2}{B-1}.$$

Then for every fixed $r \in (0, 1)$,

$$m(r) = r + \frac{V}{B-1},$$

where V is a random variable $V \sim \text{Unif}[-\frac{1}{2}, \frac{1}{2}]$. In the following, we again denote the scalar quantizer:

$$\text{quant}(t) := q_j \quad \text{when } t \in \mathcal{B}_j.$$

Then by definition,

$$\begin{aligned} \text{quant}(t) &= G\left(\frac{\lfloor (B-1)F(t) - U \rfloor + U + 1/2}{B-1}\right) \\ &= G\left(F(t) + \frac{V}{B-1}\right), \quad V \sim \text{Unif}\left[-\frac{1}{2}, \frac{1}{2}\right] \end{aligned} \quad (7)$$

Combining this with the requirements on G (Lemma A.8) gives both the unbiasedness and the scalar estimate bound.

Lemma A.9. *For every $t \in \mathbb{R}$, $\mathbb{E}_U[\text{quant}(t)] = t$. Moreover, let $X_a = \sum_{i=1}^d a_i \varepsilon_i$ where $\varepsilon_1, \dots, \varepsilon_d$ are independent Rademacher random variables and $a \in \mathbb{R}^d$ be a unit vector. Then, uniformly over $d \geq 1$ and a ,*

$$\mathbb{E}_{\varepsilon, U}(X_a - \text{quant}(X_a))^2 \leq \left(\frac{\pi\sqrt{3}}{2} + o(1)\right) B^{-2}.$$

Proof. We denote the spacing of grid buckets $\delta := \frac{1}{B-1}$. Fix $t \in \mathbb{R}$. By Eq. (5) and Eq. (7),

$$\mathbb{E}_U[\text{quant}(t)] = \frac{1}{\delta} \int_{F(t)-\delta/2}^{F(t)+\delta/2} G(s) ds = F^{-1}(F(t)) = t.$$

This proves the unbiasedness.

It remains to prove the mean squared error bound, which parallels the proof to Lemma A.1. Let $M_B := \sqrt{5 \log B}$. We first consider the contribution from $|X| \leq M_B$. By Lemma A.8, (Eq. (6)), uniformly over $|t| \leq M_B$ and $|v| \leq 1/2$,

$$G(F(t) + v\delta) = t + \frac{v\delta}{F'(t)} + o(1) \frac{\delta}{F'(t)}.$$

In our case, $v = V$ is uniform on $[-\frac{1}{2}, \frac{1}{2}]$, therefore, $\mathbb{E}V^2 = \frac{1}{12}$ and thus

$$\begin{aligned} \mathbb{E}_U(t - \text{quant}(t))^2 &= \mathbb{E}_U(t - G(F(t) + V\delta))^2 \\ &= \frac{\delta^2}{F'(t)^2} \mathbb{E}(V + o(1))^2 \\ &\leq (1 + o(1)) \frac{\delta^2}{12F'(t)^2}, \end{aligned}$$

uniformly for $|t| \leq M_B$. Now we take the expectation over X_a . Using $F'(t) = A^{-1}(2\pi)^{-1/6}e^{-t^2/6}$, $\delta = \frac{1}{B-1}$, and $\mathbb{E}_\varepsilon e^{X_a^2/3} \leq \sqrt{3}$, this becomes

$$\begin{aligned}\mathbb{E}_{\varepsilon,U} [(X_a - \text{quant}(X_a))^2 \mathbf{1}_{\{|X_a| \leq M_B\}}] &\leq (1 + o(1)) \frac{\delta^2 A^2}{12} (2\pi)^{1/3} \mathbb{E}_\varepsilon e^{X_a^2/3} \\ &= \left(\frac{\pi\sqrt{3}}{2} + o(1) \right) B^{-2}\end{aligned}$$

which concludes the central contribution. We now bound the tail contribution. Similar to Lemma A.6, we use a crude bound following from the construction of G . For every $t \in \mathbb{R}$ and every $|v| \leq 1/2$,

$$|G(F(t) + v\delta)| \leq C \left(1 + |t| + \frac{\delta}{F'(t)} \right).$$

where C is a constant. Indeed, suppose first that $t \geq 0$. In the definition of G in Lemma A.8, the values are obtained from the interval around $1/2$ by summing midpoint values of $(F^{-1})'$. Since $(F^{-1})'$ is increasing on $[1/2, 1)$, the sum up to $F(t) + v\delta$ is bounded by the corresponding integral up to $F(t)$, plus one endpoint term. This gives $C(1 + t + \delta/F'(t))$. The case $t < 0$ is symmetric, using that $F^{-1}(1-s) = -F^{-1}(s)$ and that $(F^{-1})'$ is symmetric around $1/2$.

Consequently,

$$\mathbb{E}_U(t - \text{quant}(t))^2 \leq C \left(1 + t^2 + \frac{\delta^2}{F'(t)^2} \right).$$

Applying this with $t = X_a$ gives

$$\begin{aligned}\mathbb{E}_{\varepsilon,U} [(X_a - \text{quant}(X_a))^2 \mathbf{1}_{\{|X_a| > M_B\}}] \\ \leq O \left(\mathbb{E}_\varepsilon [(1 + X_a^2) \mathbf{1}_{\{|X_a| > M_B\}}] + \mathbb{E}_\varepsilon [e^{X_a^2/3} \mathbf{1}_{\{|X_a| > M_B\}}] \right) B^{-2}.\end{aligned}$$

Using Lemma A.2 and the same argument in Lemma A.1, this tail contribution is at most

$$O(\text{poly log } B \cdot e^{-5 \log B/2}) = o(1)B^{-2}.$$

Combining the central and tail estimates, we obtain

$$\mathbb{E}_{\varepsilon,U} (X_a - \text{quant}(X_a))^2 \leq \left(\frac{\pi\sqrt{3}}{2} + o(1) \right) B^{-2}.$$

uniformly over all $d \geq 1$ and all unit vectors $a \in \mathbb{R}^d$. □

Using Lemma A.9, we immediately obtain our main theorem (Theorem 1.1), that $\tilde{x}$ is unbiased with the claimed mean squared error bounded.

Proof of Theorem 1.1. Regarding the mean squared error bound, as in the proof of 2.2, fix $x \in \mathcal{S}^{d-1}$, we have that

$$\mathbb{E}_{D,U} \|x - \tilde{x}\|_2^2 = \frac{1}{d} \sum_{i=1}^d \mathbb{E}_{D,U} (z_i - \text{quant}(z_i))^2.$$

where $z_i = \sqrt{d}y_i = \sqrt{d}(HDx)_i$ follows the distribution of X_a for some unit vector a . Applying Lemma A.9 to each coordinate gives

$$\mathbb{E}_{D,U} (z_i - \text{quant}(z_i))^2 \leq \frac{\pi\sqrt{3}/2 + o(1)}{B^2}.$$

Then averaging over i , we obtain

$$\mathbb{E}_{D,U} \|x - \tilde{x}\|_2^2 \leq \frac{\pi\sqrt{3}/2 + o(1)}{B^2}.$$

For the unbiasedness, we condition on D . Once D is fixed, the vector $y = HDx$ and hence all $z_i = \sqrt{d}y_i$ are deterministic. By the unbiasedness part of Lemma A.9,

$$\mathbb{E}_U[\text{quant}(z_i) \mid D] = z_i.$$

Thus

$$\mathbb{E}_U[\tilde{y}_i \mid D] = \frac{1}{\sqrt{d}} \mathbb{E}_U[\text{quant}(z_i) \mid D] = \frac{z_i}{\sqrt{d}} = y_i.$$

Since this holds for every coordinate,

$$\mathbb{E}_U[\tilde{y} \mid D] = y = HDx.$$

Therefore

$$\mathbb{E}_U[\tilde{x} \mid D] = DH^\top \mathbb{E}_U[\tilde{y} \mid D] = DH^\top HDx = x.$$

Finally, taking expectation over D gives $\mathbb{E}_{D,U}[\tilde{x}] = x$. $\square$

B Analysis of the inner product quantizer

We present the analysis of the inner product quantizer in this section, proving Theorem 1.2. Similarly to how the analysis of our vector quantizer relied on a moment generating function bound for Rademacher random variables, the analysis of the inner product quantizer hinges on the following fourth moment bound satisfied by Rademacher random variables:

Lemma B.1 (Mixed fourth moment). *Let $\eta_1, \dots, \eta_d$ be independent Rademacher signs. Let*

$$X = \sum_{j=1}^d a_j \eta_j, \quad Y = \sum_{j=1}^d b_j \eta_j,$$

where $\|a\| = \|b\| = 1$. Then

$$\mathbb{E}[X^2 Y^2] = 1 + 2 \langle a, b \rangle^2 - 2 \sum_{j=1}^d a_j^2 b_j^2 \leq 3.$$

Consequently,

$$\mathbb{E}[X^2(A^2 + 3Y^2)] \leq A^2 + 9.$$

Proof. Write

$$X^2 = \sum_j a_j^2 + 2 \sum_{j < k} a_j a_k \eta_j \eta_k = 1 + 2 \sum_{j < k} a_j a_k \eta_j \eta_k,$$

and similarly

$$Y^2 = 1 + 2 \sum_{j < k} b_j b_k \eta_j \eta_k.$$

Multiplying and taking expectations, all terms vanish except those with the same pair $\{j, k\}$. Therefore

$$\mathbb{E}[X^2 Y^2] = 1 + 4 \sum_{j < k} a_j a_k b_j b_k.$$

Since

$$\langle a, b \rangle^2 = \sum_j a_j^2 b_j^2 + 2 \sum_{j < k} a_j a_k b_j b_k,$$

we obtain

$$4 \sum_{j < k} a_j a_k b_j b_k = 2 \langle a, b \rangle^2 - 2 \sum_j a_j^2 b_j^2.$$

This proves the identity. The upper bound follows because $\langle a, b \rangle^2 \leq 1$ and $\sum_j a_j^2 b_j^2 \geq 0$.

Finally, since $\mathbb{E}X^2 = 1$, for any $\kappa > 1$, we have

$$\mathbb{E}[X^2(A^2 + (\kappa^2 - 1)Y^2)] \leq A^2 + 3(\kappa^2 - 1)$$

Setting $\kappa = 2$ gives the conclusion. $\square$

Roadmap. In what follows we prove an upper bound on the rate of the quantizer, i.e. show that it uses constant number of bits per coordinates, in Section B.1. We then analyze the error of the quantizer and prove Theorem 1.2 in Section B.2.

B.1 Rate Bound

In this section, we show that the output of the residual quantizer from Algorithm 3 can be stored in $O(d)$ bits.

Lemma B.2. *Let $B \geq 2$ and let $r \in \mathbb{R}^d$ such that $\|r\|_2 \leq 2$. Run the RESIDUALQUANT procedure of Algorithm 3, using Algorithm 2 for the residual scale. Then the output $\text{idx}_\sigma, \ell, \lambda$ can be stored using $(3 + \frac{1}{2\ln 2})d + O(\log \log(dB))$ total bits.*

Proof. We consider the non-trivial case $\text{idx}_\sigma > 0$. Recall that $v = HDr$. Also recall that SCALARD-EQUANT outputs $\sigma = \tau_B 2^{\text{idx}_\sigma - 1} = \tau_B 2^{\lceil \log_2(s/\tau_B) \rceil}$. By this definition, $\sigma \geq s := \frac{\|r\|_2}{\sqrt{d}}$. Therefore, if we write $Z_i := \frac{|v_i|}{\sigma}$, then

$$\sum_{i=1}^d Z_i^2 = \frac{\|HDr\|_2^2}{\sigma^2} = \frac{\|r\|_2^2}{\sigma^2} \leq d$$

where the second step follows from the fact that HD is orthogonal. Then by the definition of the level,

$$\ell_i \leq 1 + \log_2^+(Z_i),$$

where $\log_2^+(s) = \max\{0, \log_2 s\}$. Since $\log s \leq s^2/2$ for $s \geq 1$,

$$\log_2^+(Z_i) \leq \frac{Z_i^2}{2\ln 2}.$$

Thus

$$\sum_{i=1}^d (\ell_i + 1) \leq \left(2 + \frac{1}{2\ln 2}\right) d.$$

This bounds the bit usage of recording the levels $\ell_1, \dots, \ell_d$. Finally, when $\|r\|_2 \leq 2$, $s = \frac{\|r\|_2}{\sqrt{d}} \in [0, \frac{2}{\sqrt{d}}]$, thus as mentioned in Section 4, idx_σ can be encoded using

$$\lceil \log_2(\lceil \log_2(\frac{2}{\sqrt{d}}/\tau_B) \rceil + 2) \rceil = O(\log \log(dB))$$

bits. $\lambda_1, \dots, \lambda_d$ can be stored trivially in d total bits. Combining these give the claimed bound. $\square$

Remark B.3 (Expected bits per level). *Lemma B.2 gives a deterministic $O(d)$ bound on the total number of bits. Alternatively, we can also show that each level ℓ_i can be stored in $O(1)$ bits in expectation. To that effect, define*

$$Y_i = \frac{v_i}{s} = \frac{(HDr)_i}{s}.$$

The coefficient vector of Y_i has norm one, since $|H_{ij}| = d^{-1/2}$ and $s = \|r\|/\sqrt{d}$. Hence

$$\mathbb{P}(|Y_i| > t) \leq 2e^{-t^2/2}.$$

For $k \geq 1$, the event $\{\ell_i \geq k\}$ implies

$$|v_i| > \sigma 2^{k-1} \geq s 2^{k-1}.$$

Therefore

$$\mathbb{P}(\ell_i \geq k) \leq 2 \exp\left(-\frac{2^{2(k-1)}}{2}\right),$$

so the expected unary length of each level is bounded by an absolute constant.

Remark B.4 (Entropy coding the levels). *One could replace the unary encoding of the levels ℓ_i by a Huffman (or other entropy) code. This only gives a small improvement: since*

$$\Pr(\ell_i \geq k) \leq 2 \exp\left(-2^{2(k-1)}/2\right) \quad (k \geq 2),$$

the level distribution has doubly-exponential tails, and its entropy is bounded by a small absolute constant. In particular,

$$H(\ell_i) \leq \frac{2}{e \ln 2} + \sum_{k \geq 2} a_k \log_2(1/a_k) < 1.579, \quad a_k = 2 \exp\left(-2^{2(k-1)}/2\right).$$

Thus a prefix code would use, on average, fewer than $(H(\ell_i) + 1) + 1 < 3.579$ bits per coordinate including the sign bit, compared with the unary deterministic bound $3 + 1/(2 \log 2) \approx 3.72$.

B.2 Error Bounds for the Residual Quantizer

In the next lemma, we bound the error of the residual quantizer (Algorithm 3). This guarantee will later be used in the analysis of our inner product quantizer (Algorithm 4).

Lemma B.5. *Let $B \geq 2$ and let $r, y \in \mathbb{R}^d$. Run the quantization procedure of Algorithm 3, using Algorithm 2 for the residual scale. Then*

$$\mathbb{E} |\langle y, \hat{r} - r \rangle|^2 \leq 13 \frac{\|y\|^2 (\|r\|^2 + B^{-2})}{d}.$$

Proof. If $y = 0$, the claim is immediate. So we consider the non-trivial case where $y \neq 0$. Let $s = \|r\|/\sqrt{d}$. Also, let id_{σ} be the output of $\text{SCALARQUANT}(s)$ and let σ be the output of $\text{SCALARDEQUANT}(\text{id}_{\sigma})$. If $\sigma = 0$, then $\hat{r} = 0$ and $s < 1/(dB)$, so $\|r\|^2 < 1/(dB^2)$. Hence

$$|\langle y, \hat{r} - r \rangle|^2 \leq \|y\|^2 \|r\|^2 \leq \frac{\|y\|^2}{dB^2} \leq 13 \frac{\|y\|^2 (\|r\|^2 + B^{-2})}{d}.$$

Now assume $\sigma > 0$. Set $A = \frac{\sigma}{s}$. As mentioned in Section 4, by the definition of σ we have $1 \leq A < 2$. Let $u = HDy$ and $v = HDR$. Conditional on D, r , the variables $q_i - v_i$ are independent and centered, and

$$\mathbb{E} [|\langle y, \hat{r} - r \rangle|^2 \mid D, r] = \sum_{i=1}^d u_i^2 (R_i^2 - v_i^2).$$

Define

$$X_i = \frac{\sqrt{d} u_i}{\|y\|}, \quad Y_i = \frac{v_i}{s}.$$

Since $R_i = \sigma 2^{\ell_i} = As 2^{\ell_i}$ and ℓ_i is chosen to be the smallest level such that $|v_i| \leq R$, either $\ell_i = 0$ in which case $R_i = As$, or $\ell_i \geq 1$, in which case $|v_i| > As 2^{\ell_i-1}$ and multiplying by 2 gives $R_i < 2|v_i|$. In both cases, we have

$$R_i^2 - v_i^2 \leq s^2 (A^2 + (2^2 - 1)Y_i^2).$$

Therefore

$$\mathbb{E} |\langle y, \hat{r} - r \rangle|^2 \leq \frac{\|y\|^2 \|r\|^2}{d^2} \sum_{i=1}^d \mathbb{E} [X_i^2 (A^2 + 3Y_i^2)].$$

For each i , the coefficient vectors defining X_i and Y_i have norm one, so the mixed fourth-moment lemma gives

$$\mathbb{E} [X_i^2 (A^2 + 3Y_i^2)] \leq A^2 + 9.$$

Since $A < 2$, this yields

$$\mathbb{E} |\langle y, \hat{r} - r \rangle|^2 \leq 13 \frac{\|y\|^2 \|r\|^2}{d} \leq 13 \frac{\|y\|^2 (\|r\|^2 + B^{-2})}{d}.$$

□

B.3 Proof of Theorem 1.2

We now prove Theorem 1.2, restated here for convenience of the reader:

Theorem B.6 (Repeating Theorem 1.2). *Let $x \in \mathcal{S}^{d-1}$ be a unit vector and let $y \in \mathbb{R}^d$. Let $\hat{x}$ be the reconstructed vector produced by Algorithm 4 with a base bit-width b . Then*

$$\mathbb{E} |\langle y, \hat{x} - x \rangle|^2 \leq 13 \left(\frac{\pi\sqrt{3}}{2} + 1 + o(1) \right) \frac{\|y\|^2}{d \cdot 4^b},$$

where the $o(1)$ term vanishes as $b \rightarrow \infty$, uniformly over d and x .

Moreover, the output of $\text{QUANTIZE}(x)$ from Algorithm 4 can be stored using

$$db + \left(3 + \frac{1}{2 \ln 2}\right)d + O(\log(b + \log d))$$

total bits.

Proof. We write $B = 2^b$. Recall that the two-stage quantizer outputs $\hat{x} = \tilde{x} + \hat{r}$, where $r = x - \tilde{x}$. Condition on the randomness of the first stage, so that r is fixed. By Lemma B.5,

$$\mathbb{E}_{D_{\text{res}}} [|\langle y, \hat{r} - r \rangle|^2 \mid D_{\text{base}}, U] \leq 13 \frac{\|y\|^2 (\|r\|^2 + B^{-2})}{d}.$$

Taking expectation over the first stage gives

$$\mathbb{E} |\langle y, \hat{x} - x \rangle|^2 \leq 13 \frac{\|y\|^2}{d} \left(\mathbb{E}_{D_{\text{base}}, U} \|x - \tilde{x}\|_2^2 + B^{-2} \right).$$

If we use $\bar{x}$ to denote $\tilde{x}$ before projecting to the unit ball via Π , then by Theorem 2.2, we have

$$\mathbb{E}_{D_{\text{base}}, U} \|x - \bar{x}\|_2^2 \leq \frac{\pi\sqrt{3}/2 + o_B(1)}{B^2}.$$

Since the unit ball is a closed convex set and $x \in \mathcal{S}^{d-1}$, point-wise it always hold that

$$\|x - \tilde{x}\|_2^2 = \|x - \Pi\bar{x}\|_2^2 \leq \|x - \bar{x}\|_2^2$$

thus

$$\mathbb{E}_{D_{\text{base}}, U} \|x - \tilde{x}\|_2^2 \leq \frac{\pi\sqrt{3}/2 + o_B(1)}{B^2}.$$

Substituting this bound proves the theorem. The total bit usage is db bits for idx plus the bit usage from Lemma B.2. Note that we have $\|r\| \leq 2$ because after projecting onto the unit ball, both $\|x\|, \|\tilde{x}\| \leq 1$.

□